Breaking these data points down further, we really care about the amount of time a task took, so we can start by combining start time and end time into elapsed time by subtracting one from the other:

$$\text{Elapsed time } (t_{actual}) = \text{task end time} - \text{task start time}$$

Because estimates don't equal days, the second thing we'll have to do is correlate time to estimates.

Two weeks equals 10 working days, but 1 day in a sprint is used for retrospectives and planning, which gives us a total of 9 working days. To deduce time from estimates, first we'll take the highest possible estimate and equate it to the highest possible amount of time, in this case the highest possible estimate of 13 for this team equals 9 working days. Using that as our maximum, we can break down the rest of the possible estimations. To figure out the exact correlation, use the following formula:

$$max(estimate_{actual}) = (estimate_{workdays}/max(estimate_{workdays}))*max(t_{actual})$$

Once you have the maximum amount of time, you can figure out the rest of the estimate sweet spots with the following formula:

$$correlation\text{-}value = max(t_{actual})*(max(estimate_{actual})/estimate_{workdays})$$

When you know the correlation between your estimates and time, you can plug in the data you have to find out if your estimates are high or low.

Some example estimation-time mappings using these formulas are shown in table 7.1. If you're using the system outlined in appendix A, which uses EC and Lucene, I've added the queries you can use to hone in on specific data. These will become more useful when you see them in action in listing 7.1.

**Table 7.1. Mapping estimates to time ranges and validating with Lucene**

| Estimate | Exact time | Time range | Query |
|---|---|---|---|
| **Power of 2 estimations with two-week sprints** | | | |
| 16 points | 9 days | 7–9 days | devTime:[7 TO 9] AND storyPoints:16 |